

A used-car dealer gets complaints about his cars as shown in the table.

Number of complaints per day	0	1	2	3	4	5	6
Probability	0.01	0.07	0.13	0.28	0.31	0.12	0.08

Find the expected number of complaints per day.

If the random variable x can take on the n values $x_1, x_2, \dots, x_n$ with probabilities $p_1, p_2, \dots, p_n$ respectively, the expected value is $E(x) = x_1 p_1 + x_2 p_2 + \dots + x_n p_n$.

$$\begin{aligned} E(x) &= 0(0.01) + 1(0.07) + 2(0.13) + 3(0.28) + 4(0.31) + 5(0.12) + 6(0.08) \\ &= 0 + 0.07 + 0.26 + 0.84 + 1.24 + 0.60 + 0.48 \\ &= 3.49 \end{aligned}$$

The expected number of complaints per day is 3.49.